A Primitive-Direction Heat Transform for Square-Billiard Families

Route-A structural certificate C152

Abstract

We regularize the primitive direction families of the unit square billiard by a positive squared-length heat transform. Möbius inversion gives an exact theta factorization, and an elementary quarter-disk count plus Stieltjes integration gives its small-time law. Coincident lengths retain direction multiplicity. The construction is not a wave trace, isolated-orbit determinant, or target identity.

1 Transform and factorization

For ordered positive coprime pairs, coordinate swap retained and axes excluded, set

$$H_{\mathrm{prim}}(t) = \sum_{\substack{m,n\geq 1\\(m,n)=1}} e^{-4t(m^2+n^2)}, \qquad t>0. \tag{1}$$

The exponent is $-tL_{m,n}^2$ for square-billiard length $L_{m,n} = 2\sqrt{m^2+n^2}$. With $\theta_+(u) = \sum_{k\geq 1} e^{-uk^2}$, the coprimality identity gives

$$H_{\mathrm{prim}}(t) = \sum_{d\geq 1} \mu(d)\theta_+(4td^2)^2. \tag{2}$$

The interchange is absolute because monotonicity gives

$$\theta_+(4td^2) \leq \int_0^\infty e^{-4td^2x^2}\,dx = \frac{\sqrt{\pi}}{4d\sqrt{t}},$$

so the absolute transformed sum is bounded by $\frac{\pi}{16t}\sum_{d\geq 1} d^{-2}$. Each ordered primitive direction appears once, so different pairs of the same length add their multiplicities.

2 Primitive density and small time

For real $R\geq 0$, let $Q(R)$ count ordered positive lattice points in the quarter disk and let $N(R)$ count the coprime ones. Unit-square comparison gives

$$Q(R) = \frac{\pi R^2}{4} + O(R+1).$$

Möbius inversion then yields

$$N(R) = \sum_{d\leq R/\sqrt{2}} \mu(d)Q(R/d) = \frac{3R^2}{2\pi} + O(R\log R). \tag{3}$$

Indeed, absolute Dirichlet convolution and the Basel sum give $\sum_{d\geq 1}\mu(d)d^{-2} = 1/\zeta(2) = 6/\pi^2$; its omitted tail is $O(R^{-1})$, while the boundary errors sum to $O(R\sum_{d\leq R} d^{-1})$. Stieltjes integration by parts gives

$$H_{\mathrm{prim}}(t) = 8t\int_0^\infty re^{-4tr^2}N(r)\,dr.$$

Substitution of (3) gives

$$H_{\mathrm{prim}}(t) = \frac{3}{8\pi t} + O\Big(t^{-1/2}\log(1/t)\Big), \qquad t\downarrow 0. \tag{4}$$

The remainder follows directly from $t\int_0^\infty r^2\log(2+r)e^{-4tr^2}\,dr$ after the change $u=\sqrt{t}\,r$.

3 Exact validation and boundary

If b_s counts every ordered positive representation $s=a^2+b^2$ and c_s the primitive ones, (2) is coefficientwise $c_s = \sum_{d^2|s}\mu(d)b_{s/d^2}$. Producer and independent checker agree for every $s\leq 20000$ and on Euclidean radius counts through 200; SymPy independently verifies initial coefficients and the leading integral.

This is a direction-label heat transform. It is not a clean wave trace, a Gutzwiller or isolated-orbit determinant, or the spectral heat trace of the Dirichlet Laplacian.

The tuple is (`A1_WEAK`, `A2_FAIL`, `A3_FAIL`, `A4_NATURAL_QUANTIZATION`). We claim no target divisor, functional equation or counting law, arithmetic/local factor, Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.